

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 01)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

Our First Program - "Hello the WHOLE World"

- Write, Execute, and Repeat coding if needed

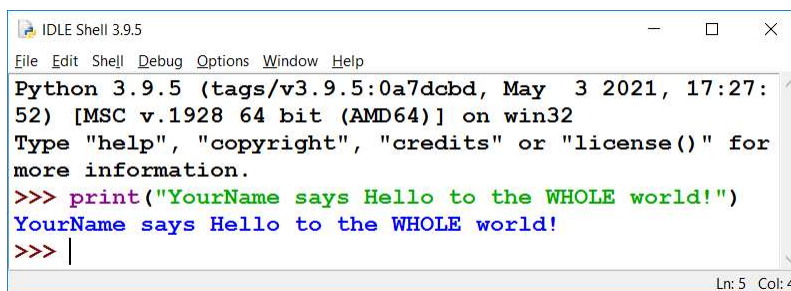
1. **Task** (Problem Requirement / Specification): write a simple Python program to

- say "Hello" to the whole world with the following sentence in the Python IDLE shell (Interactive (Script) Mode and Code Editor).

"YourName says Hello to the WHOLE world!"

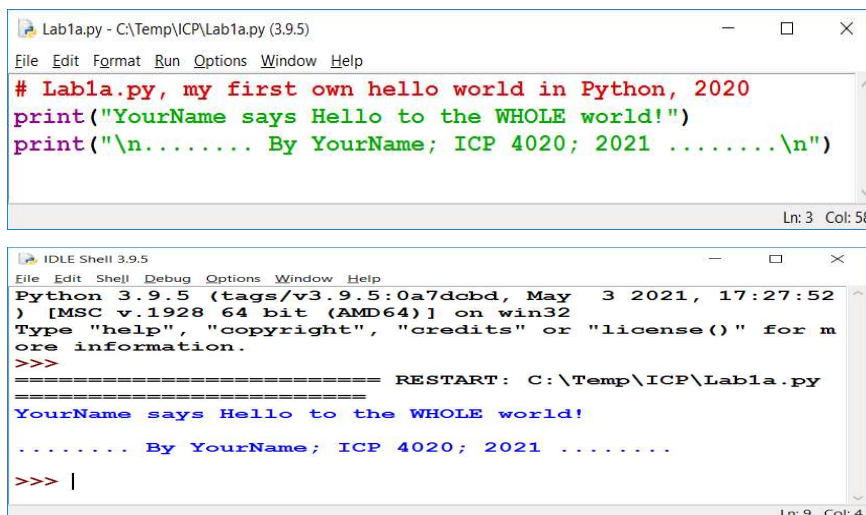
- show your actual name (e.g. **CHAN Siu Ming**), class, and year before system exits.

Interactive Mode



```
Python 3.9.5 (tags/v3.9.5:0a7dcdbd, May 3 2021, 17:27:52) [MSC v.1928 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> print("YourName says Hello to the WHOLE world!")
YourName says Hello to the WHOLE world!
>>> |
```

Source Code File Mode



```
# Lab1a.py, my first own hello world in Python, 2020
print("YourName says Hello to the WHOLE world!")
print("\n..... By YourName; ICP 4020; 2021 ..... \n")
```

```
Python 3.9.5 (tags/v3.9.5:0a7dcdbd, May 3 2021, 17:27:52) [MSC v.1928 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> ===== RESTART: C:\Temp\ICP\Lab1a.py =====
YourName says Hello to the WHOLE world!
..... By YourName; ICP 4020; 2021 .....
>>> |
```

2. *Interactive Mode*

a. *Start the programming tools:* the Python IDLE Shell (e.g. version 3.9.5)

- E.g. On Windows, from **Start > Python.. > IDLE (Python 3.9 64-bit)**

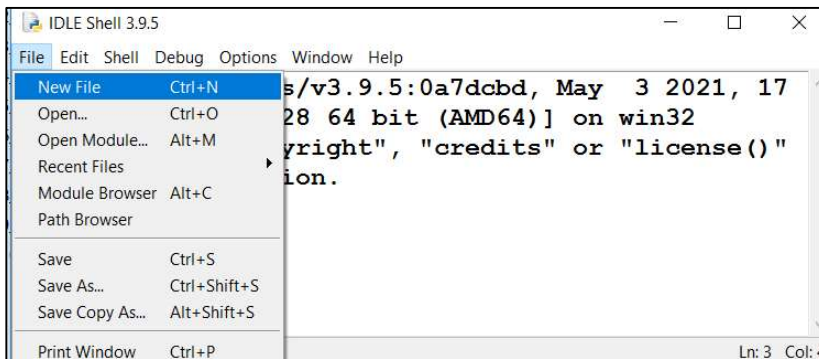
b. *Type the code line (below), ends with <ENTER> key to proceed execution for testing*

print("YourName says Hello to the WHOLE world!")

- Use your own name, e.g. **CHAN Siu Ming**

3. Source Code File Mode

- a. Start the programming tools: the **Python IDLE** Shell, if not yet.
- b. Create a new **Python (*.py)** program file with **File** > **New File** on Python Shell.



- c. Save the Python source code file as *.py, named **Lab1a.py**, to a desired folder.
- d. Type the following program codes into the space of the Code Editor of Python (Below).
 - o Use **your own name**, e.g. **CHAN Siu Ming**

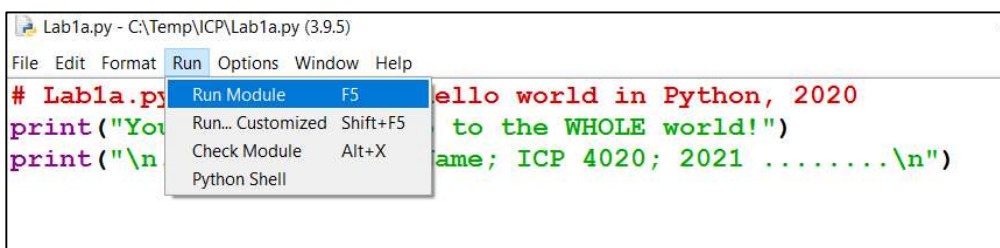
```
# Lab1a.py, my first own hello world in Python, 2021
print("YourName says Hello to the WHOLE world!")
print("\n..... By YourName; ICP 4020; 2021 ..... \n")
```

Remark:

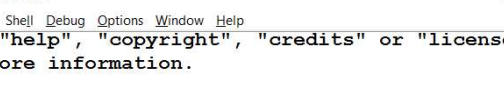
- o Symbol **#** is a read-only comment
- o Symbol **\n** is a newline escape (jump to a new next line)
- o **print()** is a display function to display characters on screen

- e. Execute the codes, with **Run** > **Run Module**, or Hotkey **<F5>**.

- o May need to save the file first, before running in the Shell



- f. Check if the program runs in **Python IDLE Shell** as expected (result printed on shell screen)



The screenshot shows an IDLE Shell window with the following content:

```
IDLE Shell 3.9.5
File Edit Shell Debug Options Window Help
Type "help", "copyright", "credits" or "license()"
for more information.
>>>
===== RESTART: C:\Temp\ICP\Lab1a.py =====
YourName says Hello to the WHOLE world!

..... By YourName; ICP 4020; 2021 .....

>>> |
```

The status bar at the bottom right indicates "Ln: 9 Col: 4".

4.
 - a. Modify your program to design your *own* face, and print/display/show it as sample below:
 - b. Save the file as **Lab1b.py**

```

YourName says Hello to the WHOLE world!
      -//-
      |   |
    .  o   <  .
    .      .
    .  _  .
    .      .
      ....

```

5. a. Write a Python program in a source file:
- First ask user to enter answer of "**What subject do you love most?** "
 - Then print and display "**ABC is a great course!**", if user enters **ABC**

b. Save the file as `Lab1c.py`

* Hint: refer to sample programs in lecture notes.

What subject do you love most? ICP
ICP is a great course!

Input
by user

AT HOME: Install Python IDLE, the free programming tool for this course

Computer laboratories of our college have been installed with some programming tools, including the Python IDLE for our course. Python IDLE is free, comparatively simple and easy to use. It serves our teaching and learning purpose well.

You should install this IDE at home. Download a copy from the following website. Installation is straight forward.

- Download and Install Python IDLE: <https://www.python.org/downloads/>
- Latest **Python** version: **3.9.6** (August 2021)

More useful information and learning resources can be found from [Beginner's Guide to Python](#) (This guide will help you to get started quickly)

- <https://wiki.python.org/moin/BeginnersGuide>
- <https://www.python.org/doc/>

Remark: Do NOT use the OLD version Python 2 in our course.